

Ali Nikkhah

Machine Learning Engineer / Researcher

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Summary

Applied ML researcher & engineer (Sharif B.Sc. Electrical Engineering 2020-2025, GPA 3.65; M.Sc. AI from Sep 2025) specialising in agentic RAG, multimodal vision-language, self-hosted LLM serving, and GPU runtime optimisation. Shipped LLM + CV pipelines at marketplace scale, self-hosted Graph RAG assistant for a national credit bureau, and BNPL risk platforms. Research focus: LLM reasoning and hallucination mitigation.

Education

M.Sc. Artificial Intelligence	Sharif University of Technology , M.Sc. Artificial Intelligence	Tehran, Iran
	• Computer Engineering Department, AI major (full-time, ongoing, no thesis yet)	Sept 2025 – present
B.Sc. Electrical Engineering	Sharif University of Technology , B.Sc. Electrical Engineering	Tehran, Iran
	• GPA: 3.65	Sept 2020 – Sept 2025
	• Thesis: Team thesis on efficient design and implementation of spread-time CDMA ASIC (my slice: FPGA implementation and pre-chip FPGA tests), supervisor Dr. Fatemeh (Noyan) Akbar, grade 20/20	
	• Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Science, Embedded Systems, Signals and Systems, Microprocessor Systems	

Experience

Iran Credit Scoring Bureau , ML Engineer — Self-Hosted Graph RAG Assistant	Tehran, Iran — Full-time · On-site
Self-hosted Graph RAG company assistant for national credit scoring — Neo4j knowledge graph, self-hosted LLM runtime on GPU, tool-calling agent.	Apr 2026 – present
• Architected self-hosted Graph RAG assistant: Neo4j knowledge graph + vector retrieval fused into an agent that answers analyst queries over credit-scoring domain data with grounded citations.	6 months
• Ran self-hosted open-weight LLMs (vLLM/TGI) on GPU; profiled and optimised GPU runtime — KV-cache, batching, quantization (AWQ/GPTQ) — cutting per-token latency and GPU memory footprint.	
• Implemented Neo4j tool-calls (Cypher generation + schema-aware retrieval) so the assistant executes graph traversals instead of hallucinating structured facts.	
• Added hallucination guardrails — retrieval grounding, citation enforcement, deterministic fallback — for a regulated financial environment.	
• Stack: Python, Neo4j, Cypher, Graph RAG, LangGraph, LangChain	
• Most recent Iran position; overlaps Sharif M.Sc. AI (Sep 2025) — full-time industry alongside full-time graduate study	

Turquoise Digital, Senior Data Scientist & Data Engineer

Tehran, Iran — Full-time · On-

site

Directed AI for regional retailers — multilingual reco, BNPL credit-risk ensemble (AUC 0.87), agentic support chatbot, centralized data lake + observability.

Sept 2025 – Mar 2026

7 months

- Centralized lake from raw CDC via PeerDB + Trino/Parquet on 5TB/day, <60s freshness; Airflow + PySpark DAGs cut memory 60%, governed by Great Expectations + Prometheus/Grafana.
- Built default-risk ensemble (XGBoost + time-series Transformers) — AUC 0.87, -22% defaults; 50+ concurrent mSPRT A/B tests with canary & auto-rollback, presented to C-suite.
- Shipped Whisper/wav2vec2.0 + LangGraph agentic chatbot — -78% ticket volume, 94% intent accuracy; deterministic fallback + policy guardrails for auditability.
- Drove portfolio allocation & risk engine (Sharpe +0.42) with streaming ClickHouse signals; LlamaIndex retrieval graph for ranking features.
- Stack: Python, PyTorch, Spark, Trino, PeerDB, Airflow
- Concurrent with Advanced Analytics Australia (Remote Contract) — 7-month overlap, split focus: Turquoise BNPL/risk/whisper vs Advanced edge CV

Advanced Analytics Australia, Computer Vision / ML Platform Engineer (Edge)

Remote · Australia (Tehran-

based) — Remote Contract

Edge CV platform across 12 retail sites, 100+ cameras @30 FPS, Kafka streaming, quantized TensorRT/TF Lite + Triton serving.

Sept 2025 – Apr 2026

8 months

- Deployed YOLO + ViT quantized (TensorRT/TF Lite) on Triton — 99.2% uptime <50ms p99, 3.5x speedup, exactly-once Kafka, 30 FPS over 100+ streams.
- Built edge MLOps with Triton model repository, zero-downtime rolling updates, automated rollback, Prometheus/Grafana per-site health.
- Cut false positives -35% via attention-guided NMS + SNR monitoring; broker architecture for 12 distributed sites.
- Stack: Python, OpenCV, YOLO, ViT, TensorRT, TFLite
- Parallel remote contract alongside Turquoise — edge CV track

Sharif University — Robust & Interpretable ML Lab, Research Collaborator — Explainable & Robust AI

Tehran, Iran — Part-time · On-

site

Robustness & interpretability harnesses for multimodal perception under distribution shift.

July 2025 – Sept 2025

3 months

- Benchmarked adversarial + natural shift suites; notebooks surfacing Captum/SHAP attributions, saliency, counterfactuals.
- Documented deploy best-practices for regulated industries with faculty.
- Stack: PyTorch, MLflow, Captum, SHAP
- Summer research sprint concurrent with industry roles

Trinity College Dublin — EmoDub Project, Machine Learning Researcher — Emotion-Aware Speech Translation

Dublin, Ireland — Part-time · Remote Collaboration

Fusion of SER transformers + wav2vec2.0 for prosody-preserving S2S translation; cross-modal acoustic+semantic transformer.

May 2025 – Nov 2025

7 months

- Trained SER front-end (transformer + wav2vec2.0 + Whisper) for affect logits; cross-modal fusion transformer for emotion-consistent decoding.
- Curated emotion-annotated corpora (EmotionNet/EmoDB) with balanced cultural representation; evaluated BLEU, Emotion F1, MOS.
- Collab. with Prof. Siobhán Clarke; reproducible codebase <https://github.com/AliNikkhah/2001/emodub-emotion-translation>.
- Stack: PyTorch, TensorFlow, Hugging Face, librosa, wav2vec2.0, Whisper
- Part-time remote research concurrent with Digikala + Turquoise + UBC

Digikala , AI Engineering R&D Lead Owned agentic RAG platform for marketplace listing intelligence — modular control pipelines (MCP), hybrid retrieval, LLM serving at 5M+ queries/day. - Led team of 5 defining R&D roadmap + SLOs; shipped MCP with LangGraph cyclic graphs, zero-downtime hot-swapping, health-checked agent composition with deterministic fallback. - Hybrid retrieval FAISS + Elasticsearch RRF + HyDE + KAG — +23% recall@10, -41% hallucination; multi-tier Redis caching, p99 <200ms. - LLM serving on vLLM (PagedAttention) + Triton — 3.2x throughput, 60% cost reduction; 200 pods, MIG/HPA/VPA, Prometheus KV-cache dashboards. - Partnered with infra/product establishing eval pipelines, A/B guardrails, canary rollouts. - Stack: Python, LangGraph, LangChain, LlamaIndex, FAISS, Elasticsearch - Full-time lead; UBC/Trinity/Sharif were part-time remote (<10h/wk)	Tehran, Iran — Full-time · On-site Jan 2025 – Aug 2025 8 months
Digikala , AI Engineer Prototyped MCP + RAG scaffolding that scaled to org-wide mandate. - Implemented LangGraph orchestration layers with health monitoring; hybrid retrieval with RRF + deterministic fallbacks for auditability. - Embedded eval telemetry, memory-efficient indexing, response caching; load-tested to 5M queries. - Stack: LangChain, LlamaIndex, FAISS, Elasticsearch, FastAPI - Promotion track — converted to R&D Lead Jan 2025	Tehran, Iran — Part-time · Hybrid Nov 2024 – Jan 2025 3 months
University of British Columbia , Research Assistant — Vision-Language Medical Imaging (Ultrasound Report Generation) ViT/BLIP + T5 for DICOM-compliant ultrasound reporting with contrastive pretraining and clinician-in-the-loop saliency. - Designed ViT/BLIP encoders + T5 generator with contrastive/self-supervised alignment; curated DICOM, de-identified QA corpora. - Deployed attention/saliency overlays for interpretability; benchmarked BLEU/ROUGE-L + clinical efficacy with radiologists; SOTA on in-house split. - Preprint in prep: Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning (2025). - Stack: PyTorch, ViT, BLIP, T5, LLaVA, CLIP - Long-running part-time research; overlapped industry roles but capped at 10h/wk — primary commitment was Digikala/Turquoise	Remote · Vancouver, Canada (Tehran-based) — Part-time · Remote Internship (10h/wk) Feb 2024 – Nov 2025 1 year 10 months
L3S Research Center — Leibniz University Hannover , Research Assistant — Video Motion Classification Texture-free motion models via optical flow + 3D CNNs + transformers; SOTA on UCF-101/HMDB-51. - Engineered optical-flow-centric datasets with temporal sampling/augmentation; improved domain generalization across lighting/texture/camera shift. - Benchmarked 3D CNN + transformer hybrids vs baselines; analyzed attention maps; reduced overfitting, SOTA accuracy. - Stack: PyTorch Lightning, OpenCV, PyAV, TensorBoard, scikit-learn, Multimodal Models - Part-time RA during B.Sc. final year — no conflict	Hannover, Germany — Part-time · Remote/On-site Apr 2023 – Feb 2024 11 months

HomaCloud, MLOps Engineer (Founding)

Built K8s-native training, CI/CD, and serving infra from scratch — Docker/K8s, Ray, MLflow, FastAPI/TF Serving, Terraform, ArgoCD.

- Provisioned multi-node K8s via Terraform; Ray + MLflow training pipelines, TF Serving + FastAPI — <10 min releases, 99.9% deploy success, 15+ models.
- GitOps with GitHub Actions, canary, spot-instance optimization; Prometheus/Grafana + ELK observability; registry + lineage.
- Stack: Kubernetes, Docker, Ray, MLflow, Terraform, ArgoCD
- During B.Sc. — clean sequential after Mapna internship

Tehran, Iran — Full-time · On-site
Aug 2021 – Feb 2022
7 months

Publications

Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning

Jan 2025

Ali Nikkhah, [Co-authors TBD], A. Supervisor
(arXiv preprint — in preparation)

Teaching

Natural Language Processing (Graduate)

TA · Sharif (Aug 2023 – Jan 2024)

Generative Models (Graduate)

TA · Sharif (Aug 2023 – Jan 2024)

Deep Learning (Graduate)

TA · Sharif (Jan 2023 – Aug 2023)

Analog Electronics

TA · Sharif (Aug 2022 – Jan 2023)

Machine Learning & Data Science

TA · Sharif (Jan 2022 – Aug 2022)

Circuit Theory & Engineering Probability

TA · Sharif (Jan 2021 – Aug 2021)

Skills

Programming: Python, C++, Go, SQL, R, Bash, TypeScript

ML & Vision: PyTorch, TensorFlow, JAX, Hugging Face, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, OpenCV, YOLO, ONNX

Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, FAISS, ChromaDB, HyDE, KAG, Graph RAG, vLLM, Triton, Neo4j, Cypher

LLM Serving & GPU: vLLM, TGI, AWQ, GPTQ, KV-cache, GPU Optimization, PagedAttention, TensorRT-LLM

Data & MLOps: Spark, Airflow, Kafka, Docker, Kubernetes, Ray, MLflow, W&B, ArgoCD, Terraform

Languages: Persian (Native), English (C2 IELTS 8.0), Arabic (Professional), French (Limited), German (Limited)